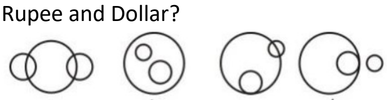
1. Select the correct word fro the options.
   After you _______, your lifestyle usually changes.
   (a) play (b) Sleep
   (c) go shopping (d) get married
2. Select the most suitable synonym:
   SWANKY
   (a) posh (b) poor
   (c) plain (d) modest
3. Which is the nearest meaning of the phrase **to rain cats and dogs?**
   (a) heavy rain (b) to rain non stop
   (c) fight among ats and dogs
   (d) drizzling
4. Fill in the blanks with the right word:
   ___________ he is well known, he is very honest
   (a) Despite the fact that
   (b) In spite of the fact that
   (c) Not with standing
   (d) Even if
5. Select the most suitable antonym:
   VAGABOND
   (a)unsettled (b) fly-by-night
   (c) aimless (d) permanent
6. Choose the correct spelling
   (a) opportunity (b) opportunity
   (c) ooportunity (d) aportonity
7. Idetify the meaning of the following idiom from the optios given.
   Spill the beans
   (a) To be undity (b) to be very talkative
   (c) to rveal something that is supposed to be kept a secret
   (d) To leave a place quickly
8. Which is the nearest meaning of the phrase **out of blue**
   (a) stranger (b) unexpectedly
   (c) to bring happiness (d) emerging out of blue colour.
9. Fill in the blank with the right option.
   A treasurer must ....the money he spends.
   (a) allow for (b) give account for
   (c) account for (b) As for
10. The phrase, A hard nut to crack means
    (a) an interesting problem
    (b) a difficult problem
    (c) a unique problem
    (d) a simple problem
11. Choose the number which is difficult from others in the group.
    (a) 8 (b) 64
    (c) 125 (d) 28
12. In a row of trees, one tree is fifth from either end of the row. How many trees are there in the row?
    (a) 8 (b) 9
    (c) 10 (d) 11
13. If – means x, x means +, + means ÷ and + means –, then
    40 x 12 + 3 – 6 + 60 = ?
    (a) 7.95 (b) 16
    (b) 44 (d) None of these
14. Choose the correct answer
    39798 + 3798 + 378 = ?
    (a)43576 (b) 43974
    (c) 43984 (d) 49532
15. Find the lowest common multiple of 24, 36 and 40.
    (a)120 (b) 240
    (c) 360 (d) 480
16. Find the number which when multiplied by 15 increased by 196.
    (a)14 (b) 20
    (c) 26 (d) 28
17. Choose the correct alternative based on relationship
    Planet : Orbit :: Projectile : ?
    (a) Trajectory (b) Track
    (c) Milky Way (d) Path
18. Choose the word which is least like the other words in the group
    (a) Pistol (b) Sword
    (c) Gun (d) Rifle
19. Choose the number which is different from others in the group.
    (a) 21 (b) 36
    (c) 49 (d) 56
20. Find out the missing number in the following sequence:
    1,3,3,6,7,9, ?, 12, 21
    (a) 10 (b) 11
    (c) 12 (d) 13
21. If in a certain code TWENTY is written as 863985 and ELEVEN is written 323039, how is TWELVE written in that code?
    (a) 863203 (b) 863584
    (c) 863903 (d) 863063
22. DGCA stands for
    (a) Director General of Civil Association
    (b) Direct General of Civil Aviation
    (c) Director General of Coast Accounts
    (d) Director General of Coal Authority
23. The 46th President of the USA is
    (a) Donald Trump (b) Joe Biden
    (c) Kamala Harris (d) Mike Pence
24. According to the NEP 2020, the school system will change to
    (a) 5+2+2+4 (b) 5+3+3+4
    (c) 5+2+3+4 (d) 5+4+3+4
25. Russian vaccine for COVID-19 is
    (a) Sputnik V (b) Sputnik VI
    (c) Sputnik (d) Sputnik IV
26. The difference between two sets {1,2,3} and {1,2,5} is the set
    (a){1} (b) {5}
    (c) {3} (d) {2}
27. The set of positive integers is
    (a) infinite (b) finite
    (c) subset (d) empty
28. Which of the following Venn diagram respresents the relation between Currency,
    Rupee and Dollar?



a b c d

    (a) a (b) b
    (c) c (d) d
29. Let n(x) denotes the number of elements in set X. If n(A) = p and n(B) = q, then how many ordered pairs (a,b) are there with a ϵ A and b ϵ B?
    (a) p^2 (b) p x q

(c) p + q (b) 2pq

30. The number of elements in the power set P(S) of the set S = [[Φ], 1,[2,3]] is
(a) 2 (b) 4
(c) 6 (d) 8

31. A set which contains a definite number of elements is called
(a) proper subset (b) universal set
(c) finite set (d) unit set

32. Determine the characteristics of the relation aRb, if $a^2 = b^2$
(a) Transitive and symmetric
(b) Reflexive and asymmetric
(c) Trichotomy, antisymmetric and irreflexive
(d) Symmetric, Reflexive and Transitive

33. In a room containing 28 people, there are 18 people who speak English, 15 people who speak Hindi and 22 people who speak Kannada, 9 people speak both English and Hindi, 11 people speak both Hindi and Kannada whereas 13 people speak both Kannada and English. How many people speak all the three languages?
(a) 6 (b) 7
(c) 8 (d) 9

34. Which of the following commands is used to delete a table in SQL?
(a) Delete (b) Drop
(c) Truncate (d) Remove

35. Which of the following standard algorithms is not a greedy algorithm?
(a) Prim's algorithm (b) Kruskal algorithm
(c) Dijkstra's shortest path algorithm
(d) Bellman-Food shortest path algorithm

36. The waterfall model of software development is
(a) The best model to use for moderate size projects
(b) the best model to use for projects with high risk of requirements change
(c) a reasonable model when requirements are well defined.
(d) a good approach when a working program is required quickly.

37. A graph is a collection of
(a) rows and columns (b) vertices and edges
(c) equations (d) matches

38. Normals distribution is also known as
(a) Cauchy's distribution (b) Laplacian distribution
(c) Gaussian distribution (d) Lagrangian distribution

39. How many distinguishable permutations of the letters are there in the word BANANA?
(a) 720 (b) 120
(c) 60 (d) 360

40. The probability of getting a job for a person is x/3. If the probability of not getting the job is 2/3, then the value of x is
(a) 2 (b) 1
(c) 3 (d) 1.5

41. If the standard deviation of x,y and z is p, then the standard deviation of 3x +5, 3y + 5, 3z + 5 is
(a) 3p + 5 (b) 3p
(c) p + 5 (d) 9p + 15

42. A man has 6 friends. In how many ways he can invite one or more of his friends at a dinner?
(a)61 (b) 62
(c) 63 (d) 64

43. Find the value of a_4 for th recurrence relation $a_n = 2a_{n-1} + 3$, with $a_0 = 6$.
(a) 320 (b) 221
(c) 141 (d) 65

44. Which of the following scheduling algorithms gives minimum average waiting time?
(a) shortest job first (b) Round robin
(c) priority (d) First come first serve

45. If for a square matrix A, $A^2 = A$, then such a matrix is known as
(a) idempotent matrix (b) orthogonal matrix
(c) null matrix (d) None of the above

46. According to the principle of mathematical induction. If P(k+1) = $m^{(k+1)}+5$ is true, then ______ must be true?
(a) $P(K) = 3m^{(k)}$ (b) $P(k) = m^{(k)} +5$
(c) $P(k)= m^{(k+2)} + 5$ (d) $P(k) = m^{(k)}$

47. The network topology supports bidirectional links between each possible node is
(a) mesh (b) star
(c) ring (d) tree

48. If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$ then
(a) $\alpha = 2ab, \beta = 2ab$
(b) $\alpha = a^2 +b^2, \beta = 2ab$
(c) $\alpha = a^2 + b^2, \beta = a^2 - b^2$
(d) $\alpha = 2ab, \beta = a^2 + b^2$

49. A CPU –bound program might have
(a) a few long CPU bursts
(b) a few short CPU bursts
(c) many short CPU bursts
(d) None of the above

50. Let us consider a square matrix, A of order n with eigen values of a,b,c. Then the eigen values of the matrix A^T could be
(a) a,b,c (b) –a, –b, –c
(c) a-b, b-a, c-a (d) a^{-1}, b^{-1}, c^{-1}

51. Which of the following operations is not O(1) for an array of sorted data? You may assume that array elements are distinct
(a)Find the ith largest element
(b) Delete an element
(c) Find the ith smallest element
(d) All of the above

52. Informal high-level description of an alrogithm in English is called
(a) function (b) class
(c) Pseudo code (d) None of the above

53. To evaluate an expression without any embedded function calls
(a) one stack is enough (b) two stacks are needed
(c) as many stacks as the height of the expression tree are needed
(d) a Turing machine is needed in the general case

54. Which of these operators is used to allocate memory to array variable in Java?
(a) Malloc (b) Alloc
(c) New (d) New malloc

55. What is the worst-case time complexity of linear search algorithm?
(a) O(n) (b) O(i)
(c) O(logn) (d) $O(n^2)$

56. Consider the following function implemented in C:
```
void test_fun(int x, int y){
int*p;
x=0;
p = &x:
y = *p:
*p =1;
printf(*%d,%d",x,y)
```

}

The output of invoking test_fun (10, 10) is

(a) 0, 0 (b) 1, 0
(c) 0, 1 (d) 1, 1

57. Which of the following data structures in a linear type?

(a) Graph (b) Tree
(c) Binary (d) Queue

58. The smallest integer than can be represented by an 8-bit number in 2's complement form is

(a)-256 (b) -128
(c) –127 (d) 0

59. Which among the following best describes polymorphism?

(a) it is the ability for a message/data to be processed in more than one form
(b) It is the ability for a message/data to be processed in only one form
(c) It is the ability for many messages/data to be processed in one way
(d) It is the ability for undefined message/data to be processed in at least one way

60. Which among the following is the language which supports classes but not polymorphism?

(a) SmallTalk (b) Ada
(c) Java (d) C++

61. ___________ is a template for entities that have common behaviour

(a) class (b) object
(c) Data (d) Polymorphism

62. Which of the following codes is used in Java to represent characters?

(a) ASCII code (b) Unicode
(c) Byte code (d) None of the above

63. Which operator is having the highest precedence?

(a) Postfix (b) Unary
(c) Shift (d) Equality

64. The given numbers are inserted into an empty binary search tree in the given order: 10,1,3,5,15,12 and 16. What is the height of the binary search tree.

(a) 3 (b) 4
(c) 5 (d) 6

65. Quicksort is a

(a) greedy algorithm (b) divide and conquer algorithm
(c) dynamic programming algorithm
(d) backtracking algorithm

66. What happens if the following program is executed in C and C++?

```
#include <stdio.h>
Int main(void)
{
 Int new = 1;
 printf ("%d", new);
}
```

(a) Error in both C and C++
(b) A successful run in both C and C++
(c) Error in C and successful execution in C++
(d) Error in C++ an successful execution in C

67. Which of the following is not a member of class?

(a) static function (b) Friend function
(c) const function (d) Virtual function

68. Which of the following is not an application of binary search?

(a) To find the lower/upper bound in an ordered sequence
(b) union of intervals
(c) Debugging
(d) To search in unordered list

69. What is the output of this C code?

```
#include<stdio.h>
Void main()
{
 Int x = 18;
 void *ptr = &x;
 printf("%f/n", *(foat)ptr);
}
```

(a)compile time error (b) undefined behaviour
(c) 10 (d) 0.000000

70. Prim's algorithm is a/an

(a) divide and conquer algorithm
(b) greedy algorithm
(c) dynamic programming
(d) approximation algorithm

71. Which of the following ways can be used to represent a graph?

(a)only Adjacency list (b) only adjacency Matrix
(c) only incidence Matrix (d) All of the above

72. In a B* tree, If the search-key value is 8 bytes long, the block size is 512 bytes and the block pointer size is 2 bytes, then maximum order of the B* tree is

(a) 50 (b) 51
(c) 52 (d) 54

73. The minimum number of functions should be present in a C++ program for its execution is

(a) 0 (b) 1
(c) 2 (d) 3

74. Given an array A = {55, 66, 77, 88, 99} and a key = 88. How many iterations are done until the element is found?

(a)1 (b) 3
(c) 4 (d) 2

75. When one object reference variable is assigned to another object reference variable, then

(a) a copy of the object is created
(b) a copy of the reference is not created
(c) it is illegal to assign one object reference variable to another object reference variable
(d)a copy of the refence is created

76. The number of different bnary trees with 6 nodes is

(a) 6 (b) 42
(c) 132 (d) 256

77. What is the scope of the variable declared in the user-defined function?

(a) in whole program (b) only inside the {} block
(c) Both 1 and 2 (d) None of these

78. Which of the following is a valid class declaration?

(a) class A {int x;} (b) class B { }
(c) public class A { } (d) object A {int x;);

79. The time factor when determining the efficiency of an algorithm is measured by

(a) counting microseconds
(b) counting the number of key operations
(c) counting the number of statements
(d) counting the kilobytes of algorithm

80. The geometrical figure shown below in flowchart represents

(a)input (b) connector
(c) decision (d) looping

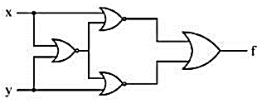
81. The complexity of binary search algorithm is
(a) O(n) (b) O(log n)
(c) $O(n^2)$ (d) O(n log n)

82. In the expression Y + X'. Y, which operator will be evaluated first?
(a) ‘ (b) +
(c) . (d) ,

83. Which of the following is false?
(a) x + y = y + x (b) x.y = y.x
(c) x.x'= 1 (d) x + x' = 1

84. A Boolean function x'y' + xy + x'y is equivalent to
(a) x' +y' (b) x + y
(c) x + y' (d) x' + y

85. Which of the following logic operations is performed by the following given combinational circuit?



(a) Exclusive-OR (b) Exclusive-NOR
(c) NAND (d) NOR

86. The output of an AND gate with three inputs A, B and C is HIGH when
(a) A = 1, B = 1, C = 0
(b) A = 0, B = 0, C = 0
(c) A = 1, B = 1, C = 1
(d) A = 1, B = 0, C = 1

87. 2's complement of 1011011 is
(a) 0100011 (b) 0110101
(c) 0100111 (d) 0100101

88. A digital circuit that can store only one bit is a/an
(a) register (b) NOR gate
(c) flip-flop (d) XOR gate

89. How much input and output needed for demultiplexer?
(a) Many inputs one output
(b) One input many outputs
(c) One input one output
(d) None of the above

90. If A, B and C are the inputs of a full adder, then the sum is given by
(a) A and B And C
(b) A Or B And C
(c) A XOR B XOR C
(d) A OR B OR C

91. The decimal equivalent of the binary number $(1011.01)_2$
(a) $(11.375)_{10}$
(b) $(10.123)_{10}$
(c) $(11.175)_{10}$
(d) $(9.23)_{10}$

92. Don't care conditions can be used for simplifying Boolean expressions in
(a) registers (b) terms
(c) K-maps (d) latches

93. A D lip-flop can be constructed from a/an ____flip-flop.
(a) S-R (b) J-K
(c) T (d) S-K

94. How many full adder and half adder are required to add 16-bit numbers?
(a) 8 half adders, 8 full adders
(b) 1 half adder, 15 full adders
(c) 16 half adders, no full adder
(d) 4 half adders, 12 full adders

95. Which flip-flops serves to be the fundamental building blocks of counters?
(a) S-R flip-flops (b) J-K flip-flops
(c) T flip-flops (d) D flip-flops

96. What is the difference between a shift-right register and a shift-left register?
(a) There is no difference
(b) The direction of the shift
(c) Propagation delay
(d) The clock input

97. Which is not characteristic of a shift register?
(a) Serieal in/parallel in
(b)Serial in/ parallel out
(c) Parallel in / serial out
(d) Parallel in / Parallel out

98. In the expression A + BC, the total numbers of minterms will be
(a) 2 (b) 3
(c) 4 (d) 5

99. Which of the following is a structured programming technique that graphically represents the detailed steps required to solve a program?
(a) Algorithm (b) Pseudocode
(c) Flowchart (d) Top-down design

100. Which of the following is a group of bits that tells the computer to perform a particular operation?
(a) Accumulator (b) Register
(c) Instruction code (d) None of these